

UBT513: CELL AND TISSUE ENGINEERING

L	T	P	Cr
3	0	0	3.0

Course Objective: This course will enable Students to understand thoroughly the key concepts of tissue organization, remodeling and strategies for restoration of tissue function. This will enable them to design tissue regeneration and tissue injury repair strategies.

Syllabus

Introduction: Basic definition, Introduction to tissue engineering, Cells as therapeutic agents with examples. Cellular fate processes, Cell differentiation, Cell migration - underlying biochemical process.

Structural and organization of tissues: Tissue organization, Tissue Components, Tissue types, Functional subunits. Tissue Dynamics, Homeostasis in highly proliferic tissues and Tissue repair. Angiogenesis. Epithelial, connective; vascularity and angiogenesis, basic wound healing, cell migration, current scope of development and use in therapeutic and in-vitro testing.

Molecular and cellular aspects: Cell-extracellular matrix interactions - Binding to the ECM, Modifying the ECM, Malfunctions in ECM signalling. Cell signalling molecules, growth factors, hormone and growth factor signalling, growth factor delivery in tissue engineering, cell attachment: differential cell adhesion, receptor-ligand binding, and Cell surface markers.

Biomaterials and scaffold: Engineering biomaterials for tissue engineering, Degradable materials (collagen, silk and polylactic acid), porosity, mechanical strength, 3-D architecture and cell incorporation. Engineering tissues for replacing bone, cartilage, tendons, ligaments, skin and liver, Bioreactors for Tissue Engineering.

Case study and regulatory issues: Case study of multiple approaches: cell transplantation and engineering for liver, musculoskeletal, cardiovascular, neural, visceral tissue engineering. Ethical, FDA and regulatory issues of tissue engineering.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Comprehend the structural organization of cells and tissues.
2. Understand the role of cell interaction, cell migration, wound healing and cellular processes.

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3. Describe the different biomaterials and its properties, design, fabrication and biomaterials selection criteria for tissue engineering scaffolds.
4. Comprehend applications of tissue engineering.

Text Books

1. Principles of tissue engineering, Robert. P.Lanza, Robert Langer & William L. Chick, Academic press, 4th edition (2014).
2. The Biomedical Engineering –Handbook, Joseph D. Bronzino, CRC press, 4th edition (2015).
3. Introduction to Biomedical Engineering, Endarle, Blanchard & Bronzino, Academic press, 3rd edition (2012).

Reference Books

1. Tissue Engineering, B. Palsson, J.A. Hubbell, R.Plonsey & J.D. Bronzino, CRC-Taylor & Francis, 1st edition (2016).
2. Nanotechnology and Tissue engineering - The Scaffold", Cato T. Laurencin, Lakshmi S. Nair, CRC Press, 1st edition (2008).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments /quizzes)	30

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